

Introduction to Biophysics II: Quantum physics & biology

Lecture 10: Energy transfer in photosynthesis : Dexter

Dexter energy transfer mechanism: exchange interaction
Dexter mechanism examples

Reading:
Modern Optical Spectroscopy, chapters 4,5,7
Additional:
Quantum Effects in Biology, chapter 3
Dexter, D.L.: A theory of sensitized luminescence in
solids. J. Chem. Phys. 21, 836–850 (1953)

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Dexter theory of energy transfer

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Dexter theory

Förster theory approximations:

- Donor and acceptor are far: no overlap between wave functions
- Dipole-dipole: point-dipole approximation
- Intermolecular interaction is small: no effect on absorption and fluorescence spectra of interacting molecules
- Spin wavefunctions are constant:
 - singlet-singlet or triplet-triplet transitions only (transition ground $\leftrightarrow$ excited state does not change the spin of the electron)

Singlet $\leftrightarrow$ triplet transition does not lead to significant energy transfer rates in widely separated molecules.

Molecules in close contact → Dexter energy transfer mechanism

Reference:

Dexter, D.L.: A theory of sensitized luminescence in solids.
J. Chem. Phys. 21, 836–850 (1953)

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Dexter energy transfer

Had before: electrons are far, considered as separate particles

Transfer between 2 states: $\Psi_1 = \psi_{1b}\psi_{2a} \rightarrow \Psi_2 = \psi_{1a}\psi_{2b}$

Born-Oppenheimer:

$$\text{Electronic part } H_{21} = \langle \phi_{1a}\phi_{2b} | \tilde{H}_{21} | \phi_{1b}\phi_{2a} \rangle \langle \chi_{1a} | \chi_{1b} \rangle \langle \chi_{2b} | \chi_{2a} \rangle$$

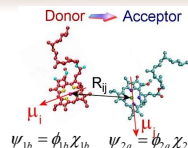
$$= H_{21(el)} \langle \chi_{1a} | \chi_{1b} \rangle \langle \chi_{2b} | \chi_{2a} \rangle.$$

$$H_{21(el)} = \langle \underbrace{\phi_{1a}\phi_{2b}}_{\Psi_2} | \tilde{H}_{21} | \underbrace{\phi_{1b}\phi_{2a}}_{\Psi_1} \rangle \approx (f^2/n^2) \int \int \phi_{1a}^* \phi_{2b}^* \frac{e^2}{r_{21}} \phi_{1b} \phi_{2a} dr_1 dr_2$$

Ψ_2
DA*

$\leftarrow$

Ψ_1
D*A

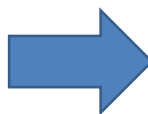


Now: electrons have overlapping wavefunctions
 → have to build mixed wave functions
 of two electrons

Expand to include spin wave functions:

Re-notate: ϕ now represent *only* spatial functions

σ represent spin wave functions:



substitute	$\phi_{1a} \rightarrow \phi_{1a}\sigma_{1a}$
	$\phi_{1b} \rightarrow \phi_{1b}\sigma_{1b}$
	$\phi_{2a} \rightarrow \phi_{2a}\sigma_{2a}$
	$\phi_{2b} \rightarrow \phi_{2b}\sigma_{2b}$

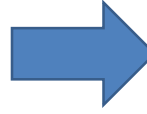
Q: How would the two-electron wavefunctions Ψ_1 and Ψ_2 change?

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Dexter energy transfer

Now: electrons have overlapping wave-functions

Expand to include *spin* wave functions:
 Re-notate: ϕ now represent *only* spatial functions
 σ represent spin wave functions:



substitute	$\phi_{1a} \rightarrow \phi_{1a}\sigma_{1a}$
	$\phi_{1b} \rightarrow \phi_{1b}\sigma_{1b}$
	$\phi_{2a} \rightarrow \phi_{2a}\sigma_{2a}$
	$\phi_{2b} \rightarrow \phi_{2b}\sigma_{2b}$

Construct joint wavefunctions: Ψ_1 – molecule 1 is excited
 Ψ_2 – molecule 2 is excited

1: molecule 1
b: excited orbital

↓

2: molecule 2
a: ground state orbital

↓

D⁺A $\Psi_1 = 2^{-1/2} \{ \underbrace{\phi_{1b}(1)\sigma_{1b}(1)\phi_{2a}(2)\sigma_{2a}(2)}_{\substack{\text{Electron 1 on excited molecule 1} \\ \text{Electron 2 in ground state of molecule 2}}} - \underbrace{\phi_{1b}(2)\sigma_{1b}(2)\phi_{2a}(1)\sigma_{2a}(1)}_{\substack{\text{Electron 2 on excited molecule 1} \\ \text{Electron 1 in ground state of molecule 2}}} \}$

D A^{*} $\Psi_2 = 2^{-1/2} \{ \underbrace{\phi_{1a}(1)\sigma_{1a}(1)\phi_{2b}(2)\sigma_{2b}(2)}_{\substack{\text{Electron 2 on excited molecule 2} \\ \text{Electron 1 in ground state of molecule 1}}} - \underbrace{\phi_{1a}(2)\sigma_{1a}(2)\phi_{2b}(1)\sigma_{2b}(1)}_{\substack{\text{Electron 1 on excited molecule 2} \\ \text{Electron 2 in ground state of molecule 1}}} \}$

(1) – coordinates of electron 1, (2) – coordinates of electron 2

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Dexter energy transfer

D⁺A $\rightarrow \Psi_1 = 2^{-1/2} \{ \phi_{1b}(1)\sigma_{1b}(1)\phi_{2a}(2)\sigma_{2a}(2) - \phi_{1b}(2)\sigma_{1b}(2)\phi_{2a}(1)\sigma_{2a}(1) \}$

D A^{*} $\rightarrow \Psi_2 = 2^{-1/2} \{ \phi_{1a}(1)\sigma_{1a}(1)\phi_{2b}(2)\sigma_{2b}(2) - \phi_{1a}(2)\sigma_{1a}(2)\phi_{2b}(1)\sigma_{2b}(1) \}$

Find matrix element: $H_{21(el)} = \langle \Psi_2 | \tilde{H}_{21(el)} | \Psi_1 \rangle$

$$\begin{aligned}
 H_{21(el)} = & \frac{1}{2} \langle \phi_{1a}(1)\sigma_{1a}(1)\phi_{2b}(2)\sigma_{2b}(2) | \tilde{H}_{21} | \phi_{1b}(1)\sigma_{1b}(1)\phi_{2a}(2)\sigma_{2a}(2) \rangle - \\
 & - \frac{1}{2} \langle \phi_{1a}(1)\sigma_{1a}(1)\phi_{2b}(2)\sigma_{2b}(2) | \tilde{H}_{21} | \phi_{1b}(2)\sigma_{1b}(2)\phi_{2a}(1)\sigma_{2a}(1) \rangle - \\
 & - \frac{1}{2} \langle \phi_{1a}(2)\sigma_{1a}(2)\phi_{2b}(1)\sigma_{2b}(1) | \tilde{H}_{21} | \phi_{1b}(1)\sigma_{1b}(1)\phi_{2a}(2)\sigma_{2a}(2) \rangle + \\
 & + \frac{1}{2} \langle \phi_{1a}(2)\sigma_{1a}(2)\phi_{2b}(1)\sigma_{2b}(1) | \tilde{H}_{21} | \phi_{1b}(2)\sigma_{1b}(2)\phi_{2a}(1)\sigma_{2a}(1) \rangle
 \end{aligned}$$

$$\begin{aligned}
 H_{21(el)} = & \frac{1}{2} \langle \phi_{1a}(1)\phi_{2b}(2) | \tilde{H}_{21} | \phi_{1b}(1)\phi_{2a}(2) \rangle \langle \sigma_{1a}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{2b}(2) | \sigma_{2a}(2) \rangle - \\
 & - \frac{1}{2} \langle \phi_{1a}(1)\phi_{2b}(2) | \tilde{H}_{21} | \phi_{1b}(2)\phi_{2a}(1) \rangle \langle \sigma_{1a}(1) | \sigma_{2a}(1) \rangle \langle \sigma_{2b}(2) | \sigma_{1b}(2) \rangle - \\
 & - \frac{1}{2} \langle \phi_{1a}(2)\phi_{2b}(1) | \tilde{H}_{21} | \phi_{1b}(1)\phi_{2a}(2) \rangle \langle \sigma_{2b}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{2a}(2) \rangle + \\
 & + \frac{1}{2} \langle \phi_{1a}(2)\phi_{2b}(1) | \tilde{H}_{21} | \phi_{1b}(2)\phi_{2a}(1) \rangle \langle \sigma_{2b}(1) | \sigma_{2a}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{1b}(2) \rangle
 \end{aligned}$$

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Dexter energy transfer

$$\begin{aligned}
 H_{21(el)} = & \frac{1}{2} \langle \phi_{1a}(1) \phi_{2b}(2) | \tilde{H}_{21} | \phi_{1b}(1) \phi_{2a}(2) \rangle \langle \sigma_{1a}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{2b}(2) | \sigma_{2a}(2) \rangle - \\
 & - \frac{1}{2} \langle \phi_{1a}(1) \phi_{2b}(2) | \tilde{H}_{21} | \phi_{1b}(2) \phi_{2a}(1) \rangle \langle \sigma_{1a}(1) | \sigma_{2a}(1) \rangle \langle \sigma_{2b}(2) | \sigma_{1b}(2) \rangle - \\
 & - \frac{1}{2} \langle \phi_{1a}(2) \phi_{2b}(1) | \tilde{H}_{21} | \phi_{1b}(1) \phi_{2a}(2) \rangle \langle \sigma_{2b}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{2a}(2) \rangle + \\
 & + \frac{1}{2} \langle \phi_{1a}(2) \phi_{2b}(1) | \tilde{H}_{21} | \phi_{1b}(2) \phi_{2a}(1) \rangle \langle \sigma_{2b}(1) | \sigma_{2a}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{1b}(2) \rangle
 \end{aligned}$$

Compare the 1st and the last terms...

$$\begin{aligned}
 & \frac{1}{2} \langle \phi_{1a}(1) \phi_{2b}(2) | \tilde{H}_{21} | \phi_{1b}(1) \phi_{2a}(2) \rangle \langle \sigma_{1a}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{2b}(2) | \sigma_{2a}(2) \rangle \\
 & \frac{1}{2} \langle \phi_{1a}(2) \phi_{2b}(1) | \tilde{H}_{21} | \phi_{1b}(2) \phi_{2a}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{1b}(2) \rangle \langle \sigma_{2b}(1) | \sigma_{2a}(1) \rangle
 \end{aligned}$$

Can swap integration variables 1 and 2 – will be the same!
 The same is true for terms 2 and 3
 Can combine terms...

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Dexter energy transfer

$$\begin{aligned}
 H_{21(el)} = & \frac{1}{2} \langle \phi_{1a}(1) \phi_{2b}(2) | \tilde{H}_{21} | \phi_{1b}(1) \phi_{2a}(2) \rangle \langle \sigma_{1a}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{2b}(2) | \sigma_{2a}(2) \rangle - \\
 & - \frac{1}{2} \langle \phi_{1a}(1) \phi_{2b}(2) | \tilde{H}_{21} | \phi_{1b}(2) \phi_{2a}(1) \rangle \langle \sigma_{1a}(1) | \sigma_{2a}(1) \rangle \langle \sigma_{2b}(2) | \sigma_{1b}(2) \rangle - \\
 & - \frac{1}{2} \langle \phi_{1a}(2) \phi_{2b}(1) | \tilde{H}_{21} | \phi_{1b}(1) \phi_{2a}(2) \rangle \langle \sigma_{2b}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{2a}(2) \rangle + \\
 & + \frac{1}{2} \langle \phi_{1a}(2) \phi_{2b}(1) | \tilde{H}_{21} | \phi_{1b}(2) \phi_{2a}(1) \rangle \langle \sigma_{2b}(1) | \sigma_{2a}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{1b}(2) \rangle
 \end{aligned}$$



Add 1st and last:

$$\langle \phi_{1a}(1) \phi_{2b}(2) | \tilde{H}_{21} | \phi_{1b}(1) \phi_{2a}(2) \rangle \langle \sigma_{1a}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{2b}(2) | \sigma_{2a}(2) \rangle$$

Compare

with Förster: $H_{21(el)} = \langle \phi_{1a} \phi_{2b} | \tilde{H}_{21} | \phi_{1b} \phi_{2a} \rangle$ in Förster spins were part of ϕ . Electrons did not change the spin: both $\langle \sigma \rangle$ are = 1

Denote it as

$$\begin{aligned}
 H_{21}^{Coulomb} &= \langle \phi_{1a}(1) \phi_{2b}(2) | \tilde{H}_{21} | \phi_{1b}(1) \phi_{2a}(2) \rangle \langle \sigma_{1a}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{2b}(2) | \sigma_{2a}(2) \rangle \\
 &= \left\langle \phi_{1a}(1) \phi_{2b}(2) \left| \frac{e^2}{r_{21}} \right| \phi_{1b}(1) \phi_{2a}(2) \right\rangle \langle \sigma_{1a}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{2b}(2) | \sigma_{2a}(2) \rangle
 \end{aligned}$$

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Dexter energy transfer

$$\begin{aligned}
 H_{21(el)} = & \frac{1}{2} \langle \phi_{1a}(1) \phi_{2b}(2) | \tilde{H}_{21} | \phi_{1b}(1) \phi_{2a}(2) \rangle \langle \sigma_{1a}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{2b}(2) | \sigma_{2a}(2) \rangle - \\
 & - \frac{1}{2} \langle \phi_{1a}(1) \phi_{2b}(2) | \tilde{H}_{21} | \phi_{1b}(2) \phi_{2a}(1) \rangle \langle \sigma_{1a}(1) | \sigma_{2a}(1) \rangle \langle \sigma_{2b}(2) | \sigma_{1b}(2) \rangle - \\
 & - \frac{1}{2} \langle \phi_{1a}(2) \phi_{2b}(1) | \tilde{H}_{21} | \phi_{1b}(1) \phi_{2a}(2) \rangle \langle \sigma_{2b}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{2a}(2) \rangle + \\
 & + \frac{1}{2} \langle \phi_{1a}(2) \phi_{2b}(1) | \tilde{H}_{21} | \phi_{1b}(2) \phi_{2a}(1) \rangle \langle \sigma_{2b}(1) | \sigma_{2a}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{1b}(2) \rangle
 \end{aligned}$$



Add 2nd and 3rd:

$$- \langle \phi_{1a}(2) \phi_{2b}(1) | \tilde{H}_{21} | \phi_{1b}(1) \phi_{2a}(2) \rangle \langle \sigma_{2b}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{2a}(2) \rangle$$

What does it represent?

Denote it as

$$\begin{aligned}
 H_{21}^{exchange} &= - \langle \phi_{1a}(2) \phi_{2b}(1) | \tilde{H}_{21} | \phi_{1b}(1) \phi_{2a}(2) \rangle \langle \sigma_{2b}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{2a}(2) \rangle \\
 &= - \langle \phi_{1a}(2) \phi_{2b}(1) | \frac{e^2}{r_{21}} | \phi_{1b}(1) \phi_{2a}(2) \rangle \langle \sigma_{2b}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{2a}(2) \rangle
 \end{aligned}$$

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Dexter energy transfer

$$D^*A \rightarrow \Psi_1 = 2^{-1/2} \{ \phi_{1b}(1) \sigma_{1b}(1) \phi_{2a}(2) \sigma_{2a}(2) - \phi_{1b}(2) \sigma_{1b}(2) \phi_{2a}(1) \sigma_{2a}(1) \}$$

$$DA^* \rightarrow \Psi_2 = 2^{-1/2} \{ \phi_{1a}(1) \sigma_{1a}(1) \phi_{2b}(2) \sigma_{2b}(2) - \phi_{1a}(2) \sigma_{1a}(2) \phi_{2b}(1) \sigma_{2b}(1) \}$$

Find matrix element: $H_{21(el)} = \langle \Psi_2 | \tilde{H}_{21(el)} | \Psi_1 \rangle$

Split into two terms: $H_{21(el)} = H_{21}^{Coulomb} + H_{21}^{exchange}$

$$\begin{aligned}
 H_{21}^{Coulomb} &= \langle \phi_{1a}(1) \phi_{2b}(2) | \tilde{H}_{21} | \phi_{1b}(1) \phi_{2a}(2) \rangle \langle \sigma_{1a}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{2b}(2) | \sigma_{2a}(2) \rangle \\
 &= \langle \phi_{1a}(1) \phi_{2b}(2) | \frac{e^2}{r_{21}} | \phi_{1b}(1) \phi_{2a}(2) \rangle \langle \sigma_{1a}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{2b}(2) | \sigma_{2a}(2) \rangle
 \end{aligned}$$

$$\begin{aligned}
 H_{21}^{exchange} &= - \langle \phi_{1a}(2) \phi_{2b}(1) | \tilde{H}_{21} | \phi_{1b}(1) \phi_{2a}(2) \rangle \langle \sigma_{2b}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{2a}(2) \rangle \\
 &= - \langle \phi_{1a}(2) \phi_{2b}(1) | \frac{e^2}{r_{21}} | \phi_{1b}(1) \phi_{2a}(2) \rangle \langle \sigma_{2b}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{2a}(2) \rangle
 \end{aligned}$$

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Coulomb term

Coulomb term: the same as in Förster theory, just a Coulomb, or *direct* interaction

$$H_{21}^{Coulomb} = \left\langle \phi_{1a}(1)\phi_{2b}(2) \left| \frac{e^2}{r_{12}} \right| \phi_{1b}(1)\phi_{2a}(2) \right\rangle \langle \sigma_{1a}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{2b}(2) | \sigma_{2a}(2) \rangle$$

$$= \left\langle \boxed{\phi_{1a}(1)\sigma_{1a}(1)} \boxed{\phi_{2b}(2)\sigma_{2b}(2)} \left| \frac{e^2}{r_{12}} \right| \boxed{\phi_{1b}(1)\sigma_{1b}(1)} \boxed{\phi_{2a}(2)\sigma_{2a}(2)} \right\rangle$$

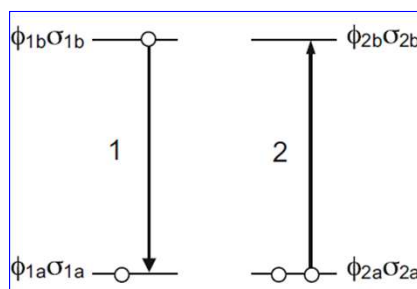
Q: Where does electron 1 go?

- From excited state of donor
to ground state of donor

Q: Where does electron 2 go?

- From ground state of acceptor
to excited state of acceptor!

Exactly as we had it for FRET



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Coulomb term

Coulomb term: the same as in Förster theory, just a Coulomb, or *direct* interaction

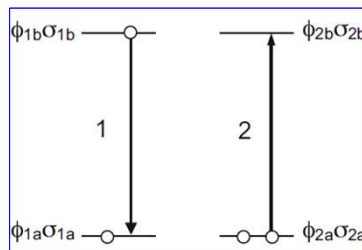
$$H_{21}^{Coulomb} = \left\langle \phi_{1a}(1)\phi_{2b}(2) \left| \frac{e^2}{r_{12}} \right| \phi_{1b}(1)\phi_{2a}(2) \right\rangle \langle \sigma_{1a}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{2b}(2) | \sigma_{2a}(2) \rangle$$

Q: What about spins during energy transfer?

$$\langle \sigma_{1a}(1) \sigma_{1b}(1) \rangle \quad \langle \alpha | \alpha \rangle = \langle \beta | \beta \rangle = 1$$

$$\langle \sigma_{2b}(2) \sigma_{2a}(2) \rangle \quad \langle \alpha | \beta \rangle = \langle \beta | \alpha \rangle = 0$$

Spins cannot change during transitions!
(for FRET)



However: one can have triplet $\rightarrow$ triplet transition or singlet $\rightarrow$ singlet transitions on the donor and acceptor in any combination, as long as spin is conserved on the donor and on the acceptor.

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Exchange term

Exchange term

$$H_{21}^{exchange} = - \left\langle \phi_{1a}(2) \phi_{2b}(1) \left| \frac{e^2}{r_{12}} \right| \phi_{1b}(1) \phi_{2a}(2) \right\rangle \langle \sigma_{2b}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{2a}(2) \rangle$$

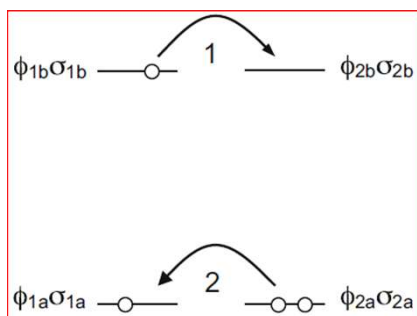
$$= - \left\langle \phi_{1a}(2) \sigma_{1a}(2) \phi_{2b}(1) \sigma_{2b}(1) \left| \frac{e^2}{r_{12}} \right| \phi_{1b}(1) \sigma_{1b}(1) \phi_{2a}(2) \sigma_{2a}(2) \right\rangle$$

Q: Where does electron 1?

- From excited state of donor
to excited state of acceptor!

Q: Where does electron 2 go?

- From ground state of acceptor
to ground state of donor!



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Exchange term: “Dexter mechanism”

Exchange term transfer mechanism (often called “Dexter energy transfer”)

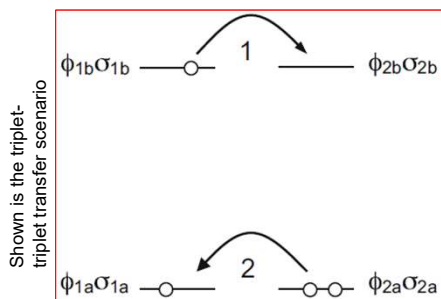
$$H_{21}^{exchange} = - \left\langle \phi_{1a}(2) \phi_{2b}(1) \left| \frac{e^2}{r_{12}} \right| \phi_{1b}(1) \phi_{2a}(2) \right\rangle \langle \sigma_{2b}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{2a}(2) \rangle$$

Q: What about spins?

The electron spins on donor
and acceptor are *interchanged*:

$$\sigma_{1b} = \sigma_{2b}$$

$$\sigma_{1a} = \sigma_{2a}$$



Note: ground and excited states may have different spins!

Exchange term allows *triplet excitation energy transfer*!

- ground state could be singlet, while excited state is triplet. Note that this is a long living state since direct transfer to ground state is spin-forbidden!

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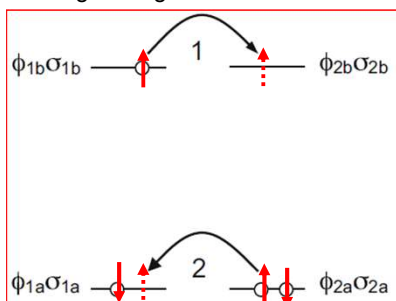
Exchange term: “Dexter mechanism”

Exchange term transfer mechanism (often called “Dexter energy transfer”)

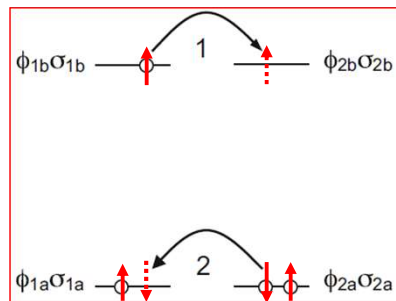
$$H_{21}^{exchange} = - \left\langle \phi_{1a}(2) \phi_{2b}(1) \left| \frac{e^2}{r_{12}} \right| \phi_{1b}(1) \phi_{2a}(2) \right\rangle \langle \sigma_{2b}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{2a}(2) \rangle$$

The electron spins on donor $\sigma_{1b} = \sigma_{2b}$
and acceptor are *interchanged*: $\sigma_{1a} = \sigma_{2a}$

Singlet-singlet transfer scenario



Triplet-triplet transfer scenario



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Exchange term: “Dexter mechanism”

Exchange term: analyzing the nature of coupling

$$H_{21}^{exchange} = - \left\langle \phi_{1a}(2) \phi_{2b}(1) \left| \frac{e^2}{r_{12}} \right| \phi_{1b}(1) \phi_{2a}(2) \right\rangle \langle \sigma_{2b}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{2a}(2) \rangle$$

Rewrite spatial term:

$$\begin{aligned} \left\langle \phi_{1a}(2) \phi_{2b}(1) \left| \frac{e^2}{r_{12}} \right| \phi_{1b}(1) \phi_{2a}(2) \right\rangle &= \int \phi_{1a}^*(\vec{r}_2) \phi_{2b}^*(\vec{r}_1) \frac{e^2}{r_{12}} \phi_{1b}(\vec{r}_1) \phi_{2a}(\vec{r}_2) d\vec{r}_1 d\vec{r}_2 \\ &= \int \phi_{2b}^*(\vec{r}_1) \phi_{1b}(\vec{r}_1) \frac{e^2}{r_{12}} \phi_{1a}^*(\vec{r}_2) \phi_{2a}(\vec{r}_2) d\vec{r}_1 d\vec{r}_2 \\ &= \int Q_1^*(\vec{r}_1) \frac{e^2}{r_{12}} Q_2(\vec{r}_2) d\vec{r}_1 d\vec{r}_2 \quad \leftarrow \text{Electrostatic Interaction of two electron clouds} \end{aligned}$$

Q: what are these “charge clouds”?

$$Q_1 \equiv \phi_{1b}^*(\vec{r}_1) \phi_{2b}(\vec{r}_1) \quad \leftarrow \text{Overlap of donor's and acceptor's excited state w-functions}$$

$$Q_2 \equiv \phi_{1a}^*(\vec{r}_2) \phi_{2a}(\vec{r}_2) \quad \leftarrow \text{Overlap of donor's and acceptor's ground state w-functions}$$

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Exchange term: “Dexter mechanism”

Exchange term:

Coulomb interaction between HOMO-HOMO and LUMO-LUMO overlaps

$$H_{21}^{exchange} = - \int Q_1^*(\vec{r}_1) \frac{e^2}{r_{12}} Q_2(\vec{r}_2) d\vec{r}_1 d\vec{r}_2 \times \langle \sigma_{2b}(1) | \sigma_{1b}(1) \rangle \langle \sigma_{1a}(2) | \sigma_{2a}(2) \rangle$$

$$Q_1 \equiv \phi_{1b}^*(\vec{r}_1) \phi_{2b}(\vec{r}_1)$$

$$Q_2 \equiv \phi_{1a}^*(\vec{r}_2) \phi_{2a}(\vec{r}_2)$$

Distance dependence:

- Wavefunctions decay exponentially away from a molecule → transfer rate also decreases exponentially with distance
- The HOMO-HOMO and LUMO-LUMO overlaps are *between* molecules, i.e. r_{12} are generally small – exchange interaction becomes large very abruptly as molecules are brought close
- When molecules are very close mutual orientation matters: overlaps could be both positive and negative and total interaction integrated over both overlaps may be zero even for very closely spaced molecules

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Example: hexatriene

Example:

Exchange interaction energy (V_T) for two hexatriene molecules separated by 3.5 Å as a function of “slip” distance

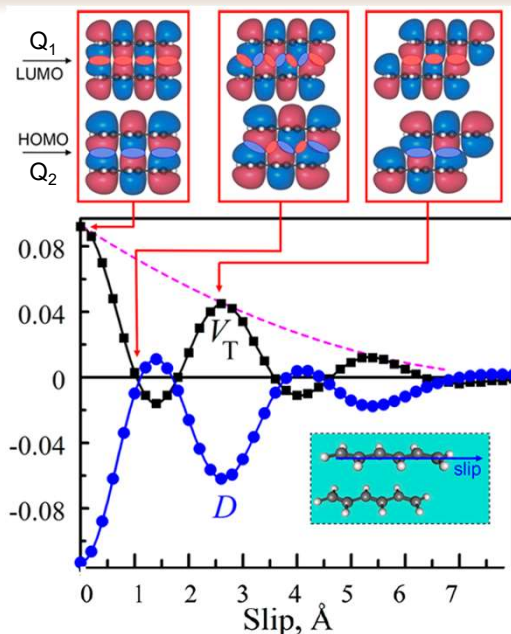
$$H_{21}^{exchange} \equiv V_T =$$

$$= - \int Q_1^*(\vec{r}_1) \frac{e^2}{r_{12}} Q_2(\vec{r}_2) d\vec{r}_1 d\vec{r}_2$$

$$Q_1 \equiv \phi_{1b}^*(\vec{r}_1) \phi_{2b}(\vec{r}_1)$$

$$Q_2 \equiv \phi_{1a}^*(\vec{r}_2) \phi_{2a}(\vec{r}_2)$$

Hartzler, Slipchenko, Savikhin, “Triplet–Triplet Coupling in Chromophore Dimers: Theory and Experiment”, Phys Chem A 2018



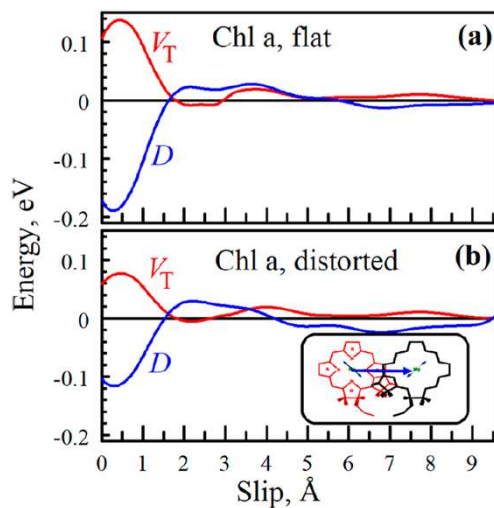
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Example: Chl *a*

Example:

Exchange interaction energy (V_T) for two Chl *a* molecules separated by 3.5 Å as a function of "slip" distance

Note: The lower graph minimizes structures, when molecules are brought together they affect the structures of each other.

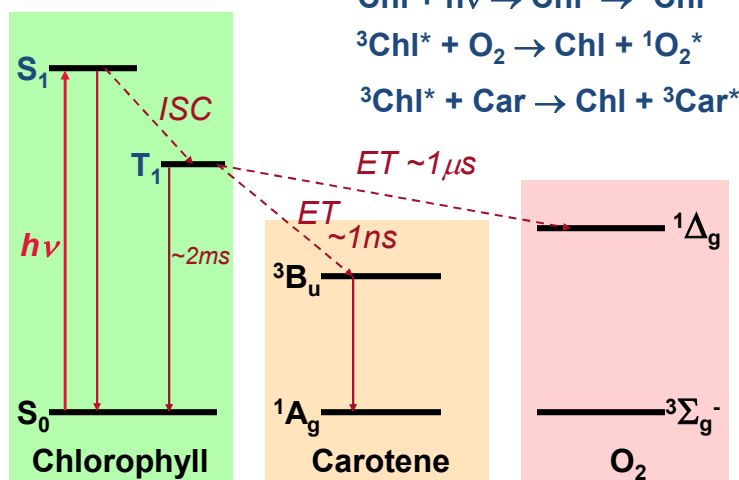


Hartzler, Slipchenko, Savikhin, "Triplet-Triplet Coupling in Chromophore Dimers: Theory and Experiment", Phys Chem A 2018

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Triplet energy transfer in photosynthesis

Triplet state can form on Chl *a* pigments!:



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Triplet energy transfer in photosynthesis

Photoprotection:

Triplet excited state $^3\text{Chl}^*$ can form from singlet excited state with rate $k \sim (10\text{ns})^{-1}$
 $^3\text{Chl}^*$ can transfer energy to molecular oxygen forming highly reactive singlet $^1\text{O}_2$
 Carotenoids are typically placed close to Chls: energy is transferred to Car which then returns to ground state (nonradiative, produces heat)

Free Chl in solution under sun light is oxidized within few minutes (in presence of oxygen).

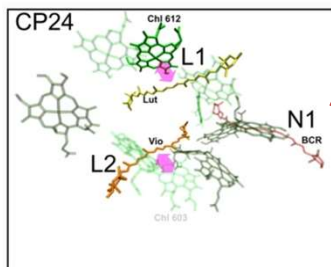
First photosynthetic organisms did not need photoprotection, since there was no oxygen in the atmosphere

~ 2.7 B years ago *cyanobacteria* invented oxygenic photosynthesis and within 500M years oxygen was rising $\rightarrow$ photoprotection evolved.

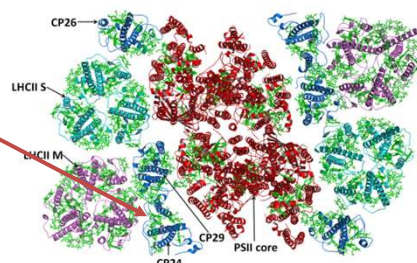
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Dexter: carotenoid $\rightarrow$ Chl transfer

Singlet Energy Transfer: can carotenoids also harvest light energy?

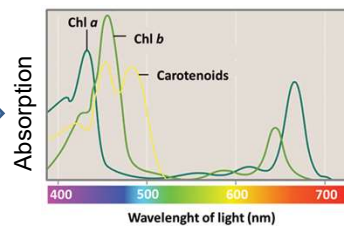


Minor antenna complex of photosystem II contains carotenoids



Photosystem II supercomplex

Problem:
 main absorption bands of carotenoids are far from Chl, no overlap!



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Dexter: carotenoid → Chl transfer

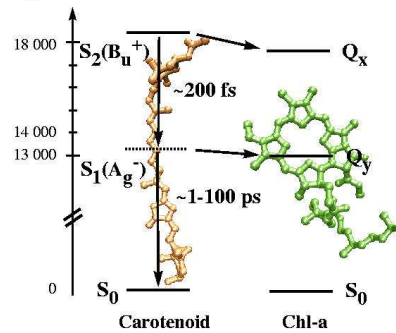
Singlet Energy Transfer: can carotenoids also harvest light energy?

Carotenoid: state with large transition moment is S_2 – it is close to Q_x band of Chl *a* but lives ~200 fs only relaxing to “dark” S_1 state that has no transition dipole moment.

If, however, close enough for Dexter mechanism (exchange coupling), energy transfer can occur from S_1 state – shown experimentally to happen.

Depending on species/structure, both energy transfers may occur.

Energy Level Scheme and Lifetimes

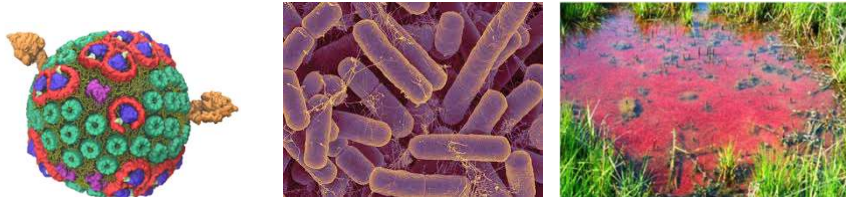


<https://www.ks.uiuc.edu/Research/psu/energy.jpg>

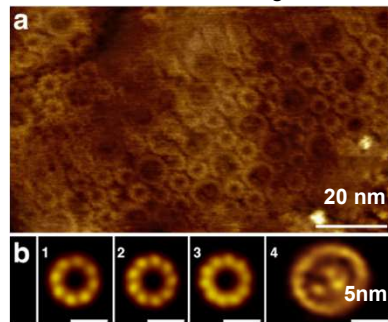
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Dexter: carotenoid → Chl transfer

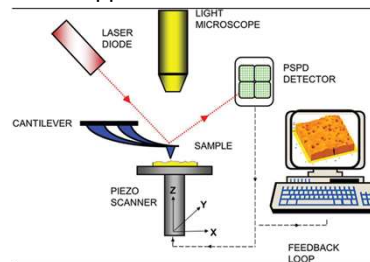
LH2 complex of photosynthetic purple sulfur bacteria



AFM images



AFM apparatus

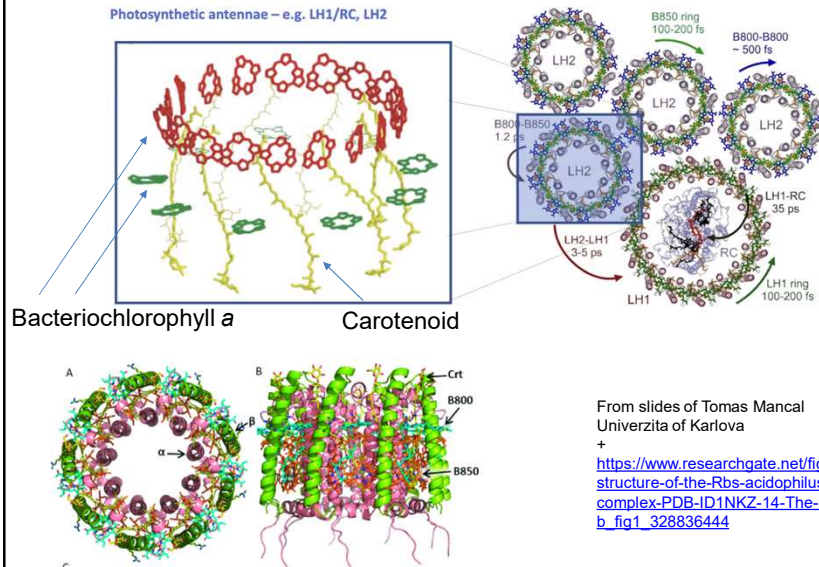


<https://alchetron.com/Purple-sulfur-bacteria>
<https://core.ac.uk/download/pdf/82342731.pdf>
<https://pharm.virginia.edu/facilities/atomic-force-microscope-afm/>
<https://www.sciencedirect.com/science/article/pii/S0092867419311717>

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Dexter: carotenoid → Chl transfer

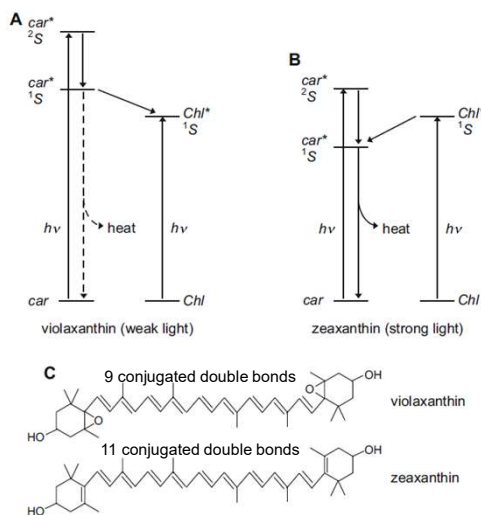
LH2 complex of photosynthetic purple sulfur bacteria



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Dexter: Chl → carotenoid transfer

Photoprotection: non-photochemical quenching



Under high light photosynthetic apparatus gets saturated by light energy leading to production of reactive species.

Plants can convert violaxanthin to zeaxanthin under low or high light intensity, respectively.

Under low light carotenoid transfers energy to Chl, and under strong light Chl transfers energy to carotenoid where it is converted to heat.

Epoxidation decreases the number of conjugated bonds in the carotenoid, which raises the energy of the lowest excited state.

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